

Rakta Bandhan — Backend Reference (App-Facing)

Scope: the donor/requester mobile app only. Admin dashboard and its Cloud Functions (`admin*` , `setAdminRole`) are a separate system and are not documented here.

1. Architecture

No custom server. The backend is **Firestore**:

- **Firestore** — the database, and the primary way the app reads/writes data directly from the client (governed by `firestore.rules`).
- **Cloud Functions** — server-side logic that can't (or shouldn't) live on the client: geo-matching, push notifications, cooldown scheduling, access-gated contact reveal. Two kinds are used:
- **Firestore triggers** (`onCreate` / `onUpdate` / `onWrite`) — fire automatically when the client writes to Firestore. This is the closest thing this backend has to a "request handler."
- **Scheduled functions** (Cloud Scheduler / pub-sub) — run on a timer, no client call involved.
- **Firestore Cloud Messaging (FCM)** — push notifications.
- **Firestore Auth** — phone-number identity; the donor/requester's `uid` is the primary key tying every collection together.

There is **no REST API** and no callable function on the app side either — every app-facing "endpoint" below is a trigger the client fires indirectly by writing to Firestore, or a schedule that runs on its own.

2. Data Model

Collection	Written by	Purpose
donors	Donor (client) on signup/update; Cloud Functions on match/donation	Private full donor profile: name, phone, blood_group, geohash, lat, lng, is_available, is_verified, fcm_token, last_seen, last_alerted_at, last_donation_date, reactivation_scheduled_at
donors_public	Cloud Function only (syncDonorPublic)	Public-safe mirror of a donor, for the nearby-donors map/list: blood_group, geohash, lat, lng, is_available, is_verified, updated_at . No name/phone.
requests	Requester (client) on create; Cloud Functions on match/expire	requester_uid, blood_group, units_needed, urgency, geohash, lat, lng, status (open/matched/fulfilled/expired/cancelled), created_at, expires_at, matched_donor_id, notified_count
notifications	Cloud Functions only	One doc per donor alerted for a request: donor_id, request_id, requester_uid, status (sent/accepted/declined/expired), distance_km, timestamp . On acceptance, also gains donor_name, donor_phone — see §4.
donation_history	Cloud Functions only	Immutable record per confirmed donation: donor_id, request_id, hospital_id, donation_date, verified_by
users	Requester (client)	App profile for non-donor users: fcm_token , etc.

3. Function Catalog (App-Facing)

Function	Trigger	Fires when	Auth	What it does
onRequestCreated	Firestore onCreate on requests/{id}	A requester creates a blood request	n/a (server-triggered)	Computes geohash bounds, queries compatible/available/verified donors within 5km (expands to 15km if none), throttles donors alerted in the last 30 min, takes the top 10 by distance, sends FCM push, writes one notifications/{id} per donor.
onDonorAccepts	Firestore onUpdate on notifications/{id}	A donor flips their notification status from sent → accepted	n/a	Transaction: locks the request (status: matched), writes donor_name / donor_phone onto that one notification

Function	Trigger	Fires when	Auth	What it does
				doc (the only place contact info is ever exposed), pushes "donor found" to the requester, expires and notifies every other donor who was alerted.
<code>onRequestCancelled</code>	Firestore <code>onUpdate</code> on <code>requests/{id}</code>	Requester (or admin) sets <code>status: cancelled</code>	n/a	Expires any still- <code>sent</code> notifications for that request and pushes a "request cancelled" alert to those donors.
<code>syncDonorPublic</code>	Firestore <code>onWrite</code> on <code>donors/{id}</code>	Any create/update/delete of a donor profile	n/a	Mirrors the non-sensitive fields into <code>donors_public/{id}</code> (or deletes the mirror if the donor doc is deleted).
<code>expireOldRequests</code>	Scheduled — every 30 min	Timer	n/a	Marks <code>open</code> requests past <code>expires_at</code> as <code>expired</code> , expires their pending notifications, notifies the requester.
<code>scheduledReactivation</code>	Scheduled — every 60 min	Timer	n/a	Re-enables donors (<code>is_available: true</code>) whose 90-day post-donation cooldown has elapsed, pushes a "you're eligible again" alert.

Not documented here (admin-only, out of scope): `onDonationConfirmed`, `setAdminRole`, `adminGetDashboardStats`, `adminVerifyDonor`, `adminToggleDonorAvailability`, `adminBanUser`, `adminManageHospital`, `adminGetAnalytics`, `adminBroadcastNotification`, `adminGetAuditLog`.

4. Security Rules Summary (`firestore.rules`)

Collection	Read	Write
<code>donors/{uid}</code>	Owner or admin only	Owner creates/updates own profile (can't self-verify); admin deletes
<code>donors_public/{uid}</code>	Any authenticated user	Cloud Function only

Collection	Read	Write
<code>requests/{id}</code>	Any authenticated user	Owner creates (must start <code>status: open</code>); owner can only cancel; Cloud Functions handle match/fulfill/expire via Admin SDK
<code>notifications/{id}</code>	The donor it's for, the requester it's for, or admin	Donor can flip <code>sent</code> → <code>accepted/declined</code> ; Cloud Functions create/expire via Admin SDK
<code>donation_history/{id}</code>	Owning donor or admin	Cloud Function only
<code>users/{uid}</code>	Owner or admin	Owner (can't unset their own ban)

Contact-reveal path: a requester never reads a donor's phone number off `donors` or `donors_public`. It only ever appears on the specific `notifications/{id}` document tied to the donor who accepted — which the rules already gate to exactly `donor_id` / `requester_uid` / admin.

5. Frontend Contract (for the app team)

- **Nearby-donors map/list screen:** query `donors_public`, not `donors`. Filter/sort client-side by `blood_group`, `geohash` range, `is_available`, `is_verified` — same fields as before, just from the public-safe collection.
- **After a donor accepts a request:** read the matched `notifications/{id}` doc (already the doc the requester is watching to know they've been matched) — it will now carry `donor_name` and `donor_phone` once accepted. Don't expect these fields on `requests` or `donors`.
- **Request lifecycle:** client only ever writes `status: open` (create) or `status: cancelled` (update-to-cancel). Every other transition (`matched`, `fulfilled`, `expired`) is server-side only.

6. Known Limitations (not fixed yet, by design for this milestone)

- **SMS fallback** — `utils/sms.js` (MSG91) is fully written but not wired into `onRequestCreated`. Push-only for now.
- **Daily alert cap** — `MAX_ALERTS_PER_DONOR_PER_DAY` is defined in `config/constants.js` but not enforced; only the 30-minute between-alerts cooldown is active.

7. Setup / Deploy (quick reference)

```
cd backend
firebase use YOUR_PROJECT_ID
cd functions && npm install
firebase deploy --only functions,firestore:rules,firestore:indexes
```

Local emulator: `firebase emulators:start` (UI at `localhost:4000`).

Full setup details: see `backend/README.md` .